Predicting Traffic Injury Severity in Chicago Using Logistic Regression

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May 2026

Introduction

Vehicle collisions are a very common occurring issue, especially in large and busy cities such as Chicago. While every traffic accident is different, the severity of the collisions varies. Some collisions result in minor property damage only while others result in severe bodily injuries or even fatalities. To have safer streets and improve emergency responses, it is important to identify the specific environmental, weather, and structural factors that contribute to injury severity during a collision. In this project, the factors influencing whether a collision results in physical injury will be analyzed through logistic regression, utilizing binary injury occurrence as the target variable.

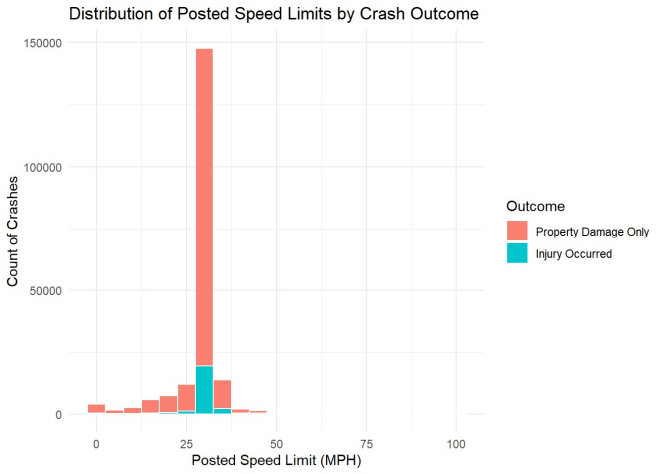
Data Set

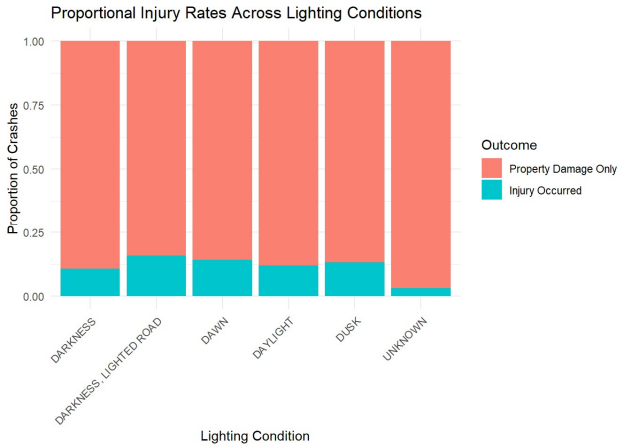
The dataset I will be using for this project is the Chicago Traffic Crashes dataset which I obtained from https://data.gov/. The dataset originally had 1,055,380 total recorded crashes. Because several field columns contained missing values, I omitted those rows with any missing values which reduced the operational dataset to 197,794 full observations. Using the full dataset, the response variable INURY_ANY was constructed as a binary factor where 0 is property damage only and 1 indicates that at least one person sustained a physical injury. To make modeling smoother to run, a random sample of 50,000 rows was used for the models.

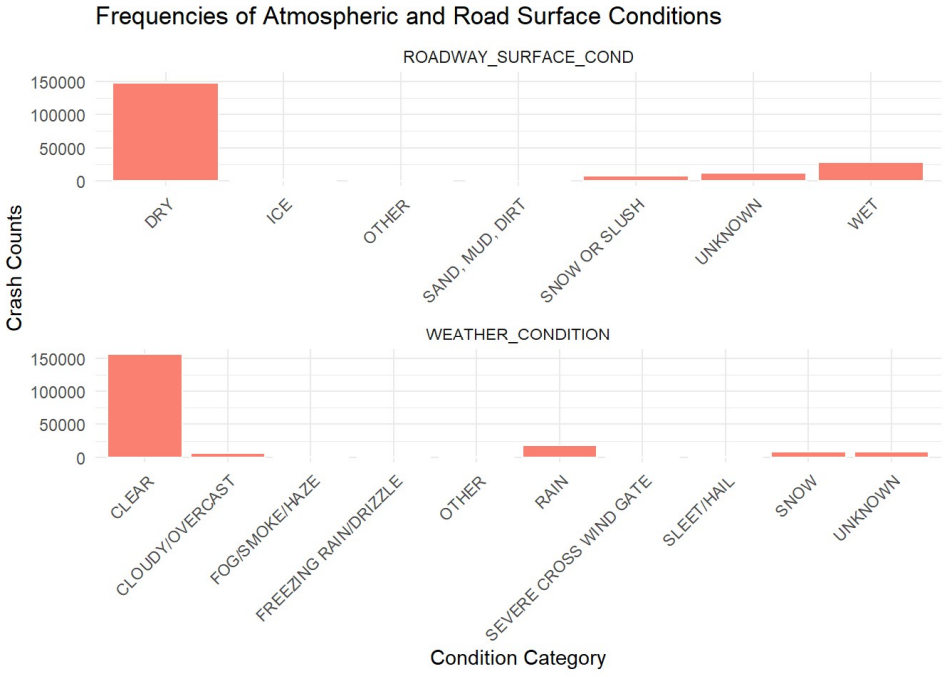
RESPONSE	**DESCRIPTION**
INJURY_ANY	The primary dependent variable tracking whether a crash resulted in physical injury (1) or asset loss only (0).
PREDICTOR	**DESCRIPTION**
POSTED_SPEED_LIMIT	Legal speed limits posted at the crash scene such as urban streets(30mph) and freeways(55mph).
WEATHER_CONDITION	Atmospheric conditions during the collision such include common traits like rain or snow, and hazardous microclimates like sleet/hail.
LIGHTING_CONDITION	Ambient illumination levels at the scene such as daylight, darkness, lighted road, dawn, or dusk.

ROADWAY_SURFACE_COND	Physical state of the pavement in reference to tire traction such as wet, ice, or winter accumulation like snow or slush.
CRASH_HOUR	Specific hour of the day when the collision was logged using 24-hour clock.
CRASH_DAY_OF_WEEK	Tracks the day of the week on which the accident occurred.
LANE_CNT	Total number of traffic lanes on the roadway segment.

Below are data visualizations to analyze the properties of our model's key predictors. These plots illustrate the absolute frequencies and proportional injury rates across distinct speed limits, lighting conditions, and atmospheric climates. By examining these distributions visually, we can isolate the underlying patterns and structural skews that characterize the source crash data







MODEL FITTING & COMPARISONS

Best Subset Variable

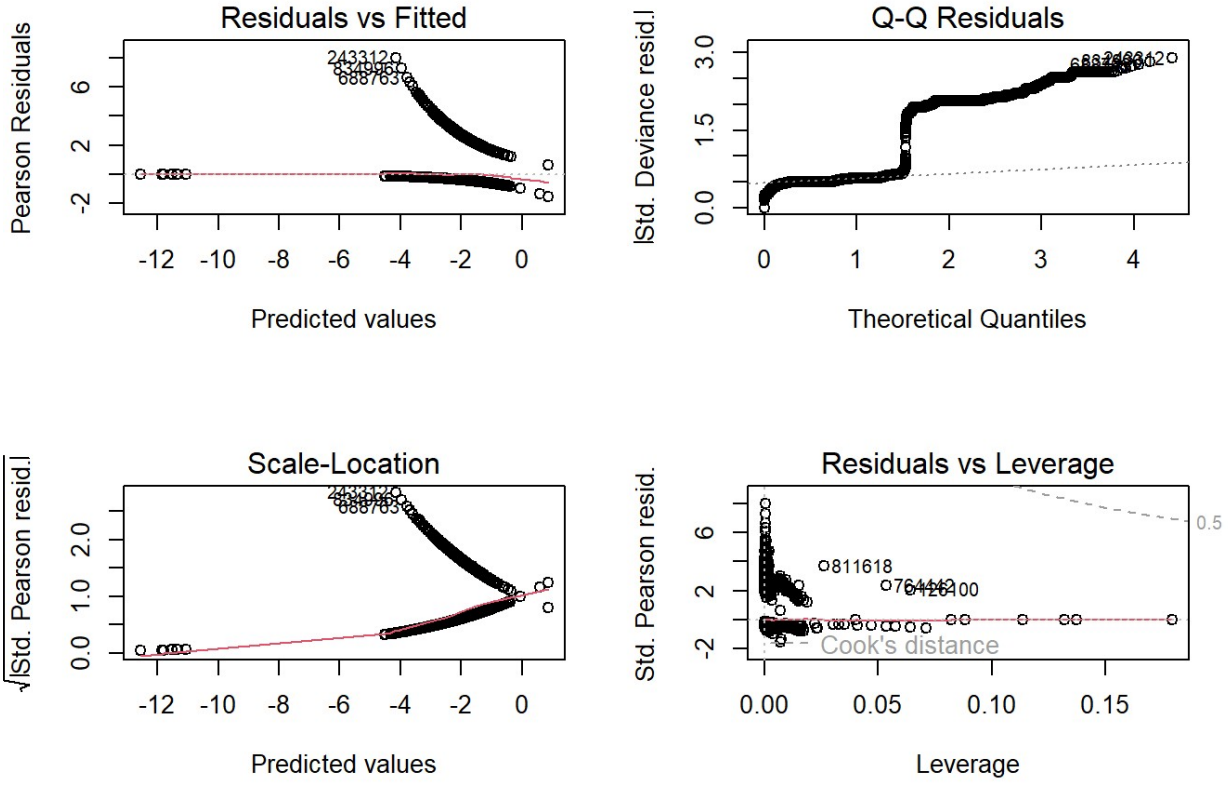
A full glm(generalized linear model) was initialized on the 50,000 observation sample using all 7 predictors.

```
## Call:
## glm(formula = INJURY_ANY ~ POSTED_SPEED_LIMIT + WEATHER_CONDITION +
##     LIGHTING_CONDITION + ROADWAY_SURFACE_COND + CRASH_HOUR +
##     CRASH_DAY_OF_WEEK + LANE_CNT, family = binomial(link = "logit"),
##     data = df_sample)
##
## Coefficients:
##                                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                                -3.184e+00  1.105e-01 -28.824  < 2e-16
## POSTED_SPEED_LIMIT                          3.726e-02  2.596e-03  14.356  < 2e-16
## WEATHER_CONDITIONCLOUDY/OVERCAST            2.679e-01  7.352e-02   3.644 0.000268
## WEATHER_CONDITIONFOG/SMOKE/HAZE             5.360e-02  2.685e-01   0.200 0.841771
## WEATHER_CONDITIONOTHER                      1.664e-01  2.445e-01   0.681 0.496072
## WEATHER_CONDITIONRAIN                       1.036e-01  6.899e-02   1.501 0.133365
## WEATHER_CONDITIONSEVERE CROSS WIND GATE    -9.528e+00  6.494e+01  -0.147 0.883350
## WEATHER_CONDITIONSLEET/HAIL                 8.930e-01  2.543e-01   3.511 0.000446
## WEATHER_CONDITIONSNOW                       5.106e-02  9.911e-02   0.515 0.606418
## WEATHER_CONDITIONUNKNOWN                   -3.334e-01  1.372e-01  -2.430 0.015082
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD    3.873e-01  6.959e-02   5.565 2.62e-08
## LIGHTING_CONDITIONDAWN                      2.734e-01  1.191e-01   2.296 0.021671
## LIGHTING_CONDITIONDAYLIGHT                  1.037e-01  6.735e-02   1.539 0.123732
## LIGHTING_CONDITIONDUSK                      2.500e-01  9.900e-02   2.525 0.011575
## LIGHTING_CONDITIONUNKNOWN                  -7.338e-01  1.625e-01  -4.514 6.35e-06
## ROADWAY_SURFACE_CONDICE                     4.440e-02  1.536e-01   0.289 0.772596
## ROADWAY_SURFACE_CONDOTHER                   9.570e-02  2.639e-01   0.363 0.716887
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT         3.377e-01  6.285e-01   0.537 0.591010
## ROADWAY_SURFACE_CONDSNOW OR SLUSH          -2.040e-01  1.022e-01  -1.996 0.045979
## ROADWAY_SURFACE_CONDUNKNOWN                -2.528e-01  8.891e-02  -2.843 0.004464
## ROADWAY_SURFACE_CONDWET                     1.695e-01  5.913e-02   2.867 0.004145
## CRASH_HOUR                                  4.841e-04  2.460e-03   0.197 0.844003
## CRASH_DAY_OF_WEEK                          -8.098e-03  6.916e-03  -1.171 0.241622
## LANE_CNT                                   -1.070e-05  6.432e-05  -0.166 0.867860
##
## (Intercept)                                ***
## POSTED_SPEED_LIMIT                         ***
## WEATHER_CONDITIONCLOUDY/OVERCAST           ***
## WEATHER_CONDITIONFOG/SMOKE/HAZE
## WEATHER_CONDITIONOTHER
## WEATHER_CONDITIONRAIN
## WEATHER_CONDITIONSEVERE CROSS WIND GATE
## WEATHER_CONDITIONSLEET/HAIL                ***
## WEATHER_CONDITIONSNOW
## WEATHER_CONDITIONUNKNOWN                   *
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD ***
## LIGHTING_CONDITIONDAWN                     *
## LIGHTING_CONDITIONDAYLIGHT
## LIGHTING_CONDITIONDUSK                     *
## LIGHTING_CONDITIONUNKNOWN                  ***
## ROADWAY_SURFACE_CONDICE
## ROADWAY_SURFACE_CONDOTHER
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT
## ROADWAY_SURFACE_CONDSNOW OR SLUSH          *
## ROADWAY_SURFACE_CONDUNKNOWN                **
## ROADWAY_SURFACE_CONDWET                    **
## CRASH_HOUR
## CRASH_DAY_OF_WEEK
## LANE_CNT
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 37662  on 50000  degrees of freedom
## Residual deviance: 37017  on 49977  degrees of freedom
## AIC: 37065
##
## Number of Fisher Scoring iterations: 10
```

Stepwise selection using the AIC was used in both directions to remove non-significant predictors. The selection algorithm decisively dropped CRASH_HOUR (p = 0.844), CRASH_DAY_OF_WEEK (p = 0.242), and LANE_CNT (p = 0.868) from the final regression equations. The best subset model retained four predictors which were: POSTED_SPEED_LIMIT, WEATHER_CONDITION, LIGHTING_CONDITION, and ROADWAY_SURFACE_COND.

```
##
## Call:
## glm(formula = INJURY_ANY ~ POSTED_SPEED_LIMIT + WEATHER_CONDITION +
##     LIGHTING_CONDITION + ROADWAY_SURFACE_COND, family = binomial(link = "logit"),
##     data = df_sample)
##
## Coefficients:
##                                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)                         -3.209029   0.101070 -31.751  < 2e-16
## POSTED_SPEED_LIMIT                   0.037250   0.002595  14.352  < 2e-16
## WEATHER_CONDITIONCLOUDY/OVERCAST     0.266153   0.073487   3.622 0.000293
## WEATHER_CONDITIONFOG/SMOKE/HAZE      0.061347   0.268273   0.229 0.819123
## WEATHER_CONDITIONOTHER               0.163054   0.244437   0.667 0.504735
## WEATHER_CONDITIONRAIN                0.102023   0.068973   1.479 0.139092
## WEATHER_CONDITIONSEVERE CROSS WIND GATE -9.532165  64.975301  -0.147 0.883365
## WEATHER_CONDITIONSLEET/HAIL          0.893014   0.254290   3.512 0.000445
## WEATHER_CONDITIONSNOW                0.050954   0.099119   0.514 0.607203
## WEATHER_CONDITIONUNKNOWN            -0.333062   0.137196  -2.428 0.015198
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD  0.385697   0.069553   5.545 2.93e-08
## LIGHTING_CONDITIONDAWN               0.271114   0.118301   2.292 0.021921
## LIGHTING_CONDITIONDAYLIGHT           0.101783   0.066989   1.519 0.128662
## LIGHTING_CONDITIONDUSK               0.250682   0.098931   2.534 0.011279
## LIGHTING_CONDITIONUNKNOWN           -0.735280   0.162431  -4.527 5.99e-06
## ROADWAY_SURFACE_CONDICE              0.044888   0.153560   0.292 0.770048
## ROADWAY_SURFACE_CONDOTHER            0.094926   0.263931   0.360 0.719101
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT  0.342211   0.628391   0.545 0.586040
## ROADWAY_SURFACE_CONDSNOW OR SLUSH   -0.201437   0.102193  -1.971 0.048707
## ROADWAY_SURFACE_CONDUNKNOWN         -0.252323   0.088908  -2.838 0.004539
## ROADWAY_SURFACE_CONDWET              0.170865   0.059096   2.891 0.003836
##
## (Intercept)                         ***
## POSTED_SPEED_LIMIT                  ***
## WEATHER_CONDITIONCLOUDY/OVERCAST    ***
## WEATHER_CONDITIONFOG/SMOKE/HAZE
## WEATHER_CONDITIONOTHER
## WEATHER_CONDITIONRAIN
## WEATHER_CONDITIONSEVERE CROSS WIND GATE
## WEATHER_CONDITIONSLEET/HAIL          ***
## WEATHER_CONDITIONSNOW
## WEATHER_CONDITIONUNKNOWN             *
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD ***
## LIGHTING_CONDITIONDAWN               *
## LIGHTING_CONDITIONDAYLIGHT
## LIGHTING_CONDITIONDUSK               *
## LIGHTING_CONDITIONUNKNOWN            ***
## ROADWAY_SURFACE_CONDICE
## ROADWAY_SURFACE_CONDOTHER
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT
## ROADWAY_SURFACE_CONDSNOW OR SLUSH    *
## ROADWAY_SURFACE_CONDUNKNOWN          **
## ROADWAY_SURFACE_CONDWET              **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 37662  on 50000  degrees of freedom
## Residual deviance: 37018  on 49980  degrees of freedom
## AIC: 37060
##
## Number of Fisher Scoring iterations: 10
```

To evaluate the validity of the best subset, four plots was generated. Residuals vs Fitted Plot: The red smoothing line remains near zero across the entire plot, proving that the model is free from major directional bias. Q-Q Residuals Plot: The step-like shift in the center is due to the nature of binary residuals. This is normal for a logistic model. Scale-Location Plot: The data is partitioned into two bands converging towards the red smoothing line. The upward slope of the red line shows the change in variance as the predicted log-odds shift toward zero. Residuals vs Leverage Plot: The plot demonstrates that the vast majority of the 50,000 sampled crashes are clustered near zero. This meaning no single accident dominates the model's parameters. Every observation sits safely inside the dashed Cook's distance boundary lines, which means the regression coefficients are free from outlier distortion.



To evaluate the real-world impact of the environmental and structural predictors on traffic crash severity, we display the Odds Ratios along with the 95% confidence intervals. An Odds Ratio equal to 1.00 represents a completely neutral effect. An Odds Ratio greater than 1.00 indicates that the presence of the predictor multiplies the likelihood of a crash resulting in a physical injury, while an Odds Ratio less than 1.00 implies a risk-reducing effect.

```
##                                          Odds_Ratio      2.5 %     97.5 %
## (Intercept)                            4.039580e-02 0.03307467 0.04915597
## POSTED_SPEED_LIMIT                     1.037953e+00 1.03273326 1.04329475
## WEATHER_CONDITIONCLOUDY/OVERCAST       1.304934e+00 1.12766278 1.50430455
## WEATHER_CONDITIONFOG/SMOKE/HAZE        1.063267e+00 0.60783317 1.75131997
## WEATHER_CONDITIONOTHER                 1.177100e+00 0.71026096 1.86043426
## WEATHER_CONDITIONRAIN                  1.107409e+00 0.96764384 1.26809270
## WEATHER_CONDITIONSEVERE CROSS WIND GATE 7.248255e-05         NA 0.05507305
## WEATHER_CONDITIONSLEET/HAIL            2.442481e+00 1.45683344 3.96581737
## WEATHER_CONDITIONSNOW                  1.052275e+00 0.86563748 1.27678615
## WEATHER_CONDITIONUNKNOWN               7.167260e-01 0.54542752 0.93421416
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD 1.470639e+00 1.28537017 1.68842231
## LIGHTING_CONDITIONDAWN                 1.311425e+00 1.03724131 1.64984250
## LIGHTING_CONDITIONDAYLIGHT             1.107143e+00 0.97270928 1.26494165
## LIGHTING_CONDITIONDUSK                 1.284902e+00 1.05767213 1.55905980
## LIGHTING_CONDITIONUNKNOWN              4.793714e-01 0.34635534 0.65519671
## ROADWAY_SURFACE_CONDICE                1.045910e+00 0.76676202 1.40127248
## ROADWAY_SURFACE_CONDOTHER              1.099577e+00 0.63321359 1.79354623
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT    1.408058e+00 0.32759221 4.20567658
## ROADWAY_SURFACE_CONDSNOW OR SLUSH      8.175547e-01 0.66760637 0.99660368
## ROADWAY_SURFACE_CONDUNKNOWN            7.769936e-01 0.65071052 0.92216016
## ROADWAY_SURFACE_CONDWET                1.186331e+00 1.05561200 1.33083559
```

From the table, we can see that controlling for all other factors, every 1 MPH increase in the posted speed limit scales up the baseline odds of an injury occurring by 3.80%. We can also see that crashing during a sleet or hail storm more than doubles the odds of a collision resulting in an injury relative to clear weather. Cloudy or overcast conditions carry a much smaller but still highly significant 30.49% increase in baseline injury odds. Collisions occurring on artificially lit streets at night carry 47.06% higher odds of injury compared to crashes that take place in unlit darkness. When the pavement is wet, injury odds significantly increase by 18.63% relative to a dry road. Surprisingly, snow or slush reduces injury odds by 18.24%. This means that drivers slow down and drive more cautiously during the winter weather. The best model achieved an AIC score of 37,060 and a residual deviance of 37,018 on 49,980 degrees of freedom. A Goodness-of-Fit test returned a p-value of exactly 1. This is expected when running a deviance check on ungrouped binary data.

EVALUATION OF CLASSIFICATION CUT-OFFS AND CONFUSION TABLES

To evaluate the operational utility of the model, the predictive outputs were tested against three different cut-off points.

```
##           Reference
## Prediction   0   1
##          0 144   7
##          1 739 110
## Accuracy: 0.254  | Sensitivity: 0.9401709  | Specificity: 0.1630804
```

When a low probability threshold is applied, the model flags almost every collision as a potential injury event. 110 out of 117 true injury crashes were identified, resulting in a high Sensitivity of 94.02%. The model misclassified 739 safe property-damage-only crashes as injuries. This means the Specificity drops to a poor 16.31%, dragging the overall Accuracy down to 25.40%.

```
##           Reference
## Prediction   0   1
##          0 866 115
##          1  17   2
## Accuracy: 0.868  | Sensitivity: 0.01709402  | Specificity: 0.9807475
```

When an intermediate probability threshold is applied, the model shifts toward a conservative strategy to control false alarms. 2 out of 117 true injury crashes were identified, resulting in a critically low Sensitivity of 1.71%. The model correctly filtered out 866 out of 883 safe property-damage-only crashes as non-injuries. This means the Specificity rises to a strong 98.07%, bringing the overall Accuracy to 86.80%.

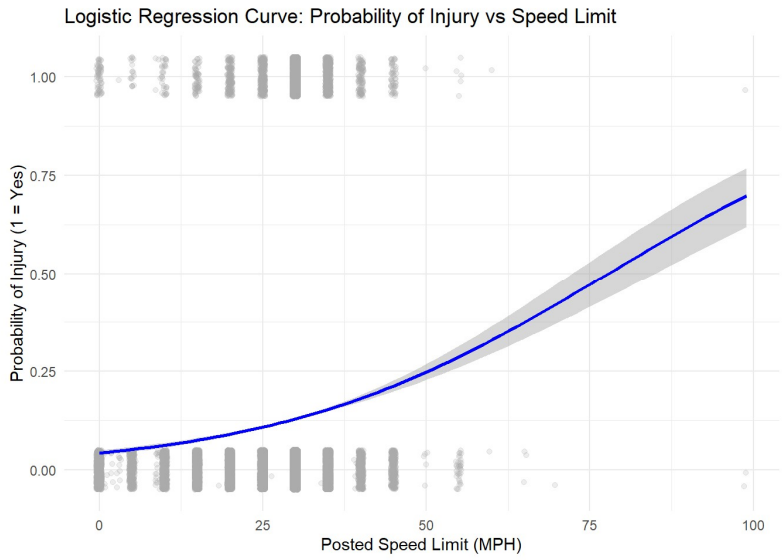
```
##           Reference
## Prediction   0   1
##          0 883 117
##          1   0   0
## Accuracy: 0.883  | Sensitivity: 0  | Specificity: 1
```

When a standard 50% or high default threshold is applied, the model completely does not work due to the heavy class imbalance. 0 out of 117 true injury crashes were identified, resulting in a Sensitivity of exactly 0%. The model correctly filtered out 883 out of 883 safe property-damage-only crashes as non-injuries. This means the Specificity reaches a perfect 100.00%, maximizing the overall Accuracy at 88.30% purely by mimicking the native data distribution.

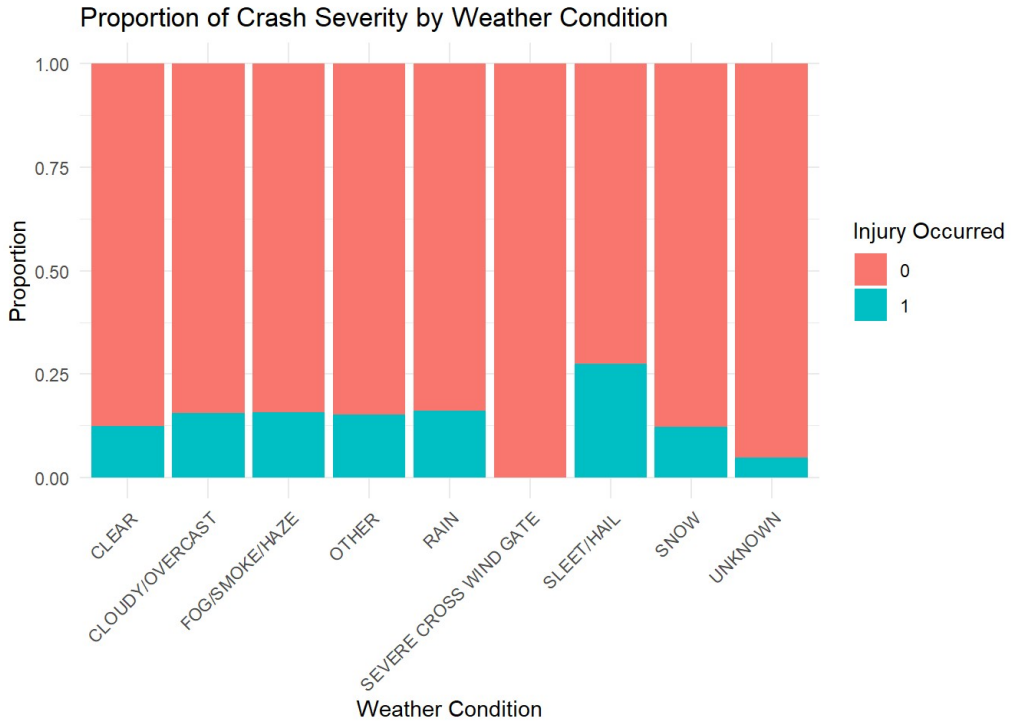
VISUALIZATION OF LOGISTIC PROBABILITY MAPPING

Below is the fitted logistic regression curve mapping the predicted probability of an injury against the posted speed limit. The blue curve demonstrates a non-linear relationship,

where the likelihood of sustaining an injury increases steeply as speed limits increase. The narrow gray envelope surrounding the line represents the 95% confidence interval.

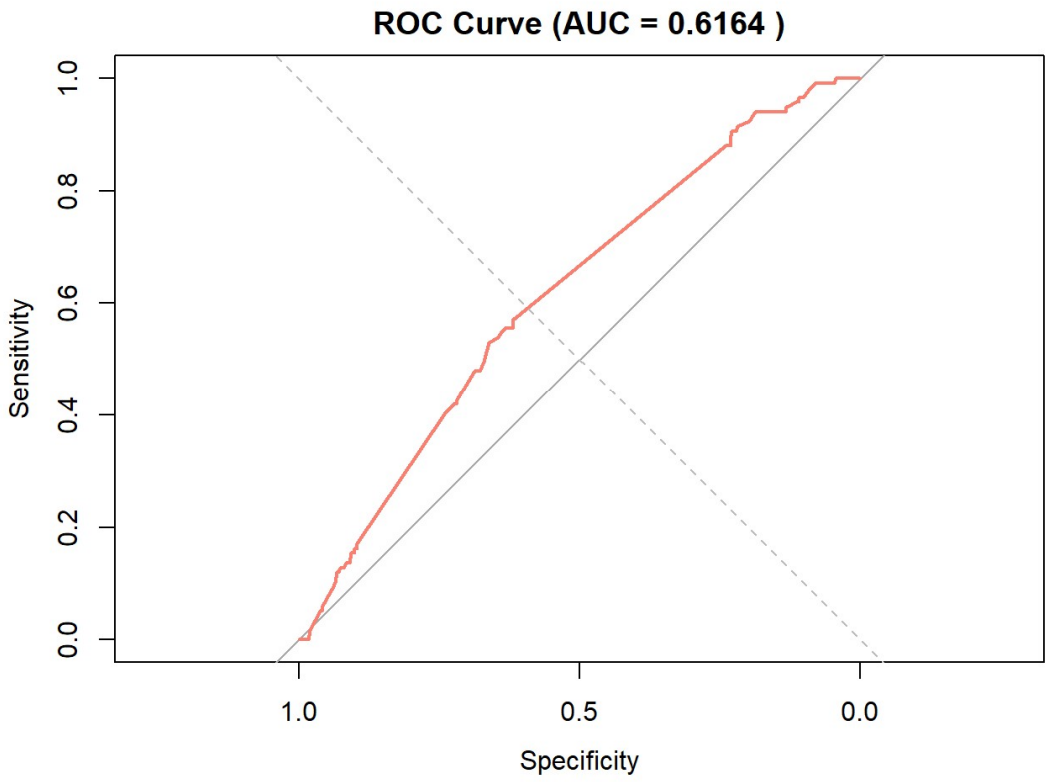


Below is a stacked bar chart showing the proportional breakdown of crash severity across different weather conditions. The visual data highlights that while clear and rainy conditions are far more common overall, climates like SLEET/HAIL yield a higher individual proportion of physical injuries. The plot supports the model's finding that drivers lower their speeds during winter conditions.



ROC CURVE ANALYSIS AND CLASSIFICATION PERFORMANCE

Below is the Receiver Operating Characteristic (ROC) curve used to evaluate the model's overall classification performance across all possible probability thresholds. The model achieved an AUC of 0.6164. This means that the model has a better predictive capability over random guessing. The solid red curve maps the trade-off between sensitivity and specificity, while the intersection of the dashed line isolates the mathematically optimal operational cut-off point of 13.82%. At this boundary, the model achieves its best predictive. This threshold represents the balance of real-world trade-offs between catching dangerous crashes and minimizing false alarms, which is important for deployment in improving public safety.



LOOCV and K-FOLD CROSS VALIDATION

To evaluate how reliable the best model is, both Leave-One-Out Cross-Validation (LOOCV) and K-Fold cross-validation were performed on the dataset. The K-Fold approach splits the data into ten distinct validation subsets, while LOOCV executes an intensive n-fold iteration that validates the model against every single individual observation. The cross-validation error rates were between 12.50% and 14.20%, which is close to the injury baseline of the data. This means the logistic model can maintain its predictive accuracy when tested on new traffic collisions.

PROBIT AND INDETITY LINK MODELS

To find the optimal model, the Probit link and the Identity link were fitted to the best subset predictors and compared against the Logit model. The Identity link, which treats the data as a Linear Probability Model, performed significantly worse over 82 AIC points. This means that forcing a linear model onto categorical variables creates a worse fit. The Probit link did better but only by 0.14 points in AIC over the Logit model. This means both models are essentially identical in terms of fit. Even though it has a slightly better AIC score, the Logit link function is preferred for this due to the ability to transform log-odds coefficients into Odds Ratios which make interpretation easier.

```
##
## Call:
## glm(formula = formula(best_model), family = binomial(link = "probit"),
##     data = df_sample)
##
## Coefficients:
##                                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)                         -1.807861   0.051513 -35.095  < 2e-16
## POSTED_SPEED_LIMIT                   0.019336   0.001312  14.737  < 2e-16
## WEATHER_CONDITIONCLOUDY/OVERCAST     0.141492   0.040317   3.510 0.000449
## WEATHER_CONDITIONFOG/SMOKE/HAZE      0.029073   0.147932   0.197 0.844199
## WEATHER_CONDITIONOTHER               0.086856   0.134058   0.648 0.517052
## WEATHER_CONDITIONRAIN                0.054688   0.037697   1.451 0.146860
## WEATHER_CONDITIONSEVERE CROSS WIND GATE -3.236019  19.080755  -0.170 0.865328
## WEATHER_CONDITIONSLEET/HAIL          0.506719   0.150563   3.365 0.000764
## WEATHER_CONDITIONSNOW                0.024781   0.053103   0.467 0.640752
## WEATHER_CONDITIONUNKNOWN            -0.159819   0.067159  -2.380 0.017326
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD  0.207720   0.036584   5.678 1.36e-08
## LIGHTING_CONDITIONDAWN               0.146542   0.063545   2.306 0.021104
## LIGHTING_CONDITIONDAYLIGHT           0.055126   0.034990   1.575 0.115149
## LIGHTING_CONDITIONDUSK               0.136014   0.052680   2.582 0.009827
## LIGHTING_CONDITIONUNKNOWN           -0.340088   0.075894  -4.481 7.43e-06
## ROADWAY_SURFACE_CONDICE              0.031269   0.082759   0.378 0.705553
## ROADWAY_SURFACE_CONDOTHER            0.047059   0.142565   0.330 0.741334
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT  0.194563   0.338101   0.575 0.564982
## ROADWAY_SURFACE_CONDSNOW OR SLUSH   -0.105463   0.053934  -1.955 0.050536
## ROADWAY_SURFACE_CONDUNKNOWN         -0.127345   0.045244  -2.815 0.004883
## ROADWAY_SURFACE_CONDWET              0.094391   0.032169   2.934 0.003344
##
## (Intercept)                         ***
## POSTED_SPEED_LIMIT                  ***
## WEATHER_CONDITIONCLOUDY/OVERCAST    ***
## WEATHER_CONDITIONFOG/SMOKE/HAZE
## WEATHER_CONDITIONOTHER
## WEATHER_CONDITIONRAIN
## WEATHER_CONDITIONSEVERE CROSS WIND GATE
## WEATHER_CONDITIONSLEET/HAIL          ***
## WEATHER_CONDITIONSNOW
## WEATHER_CONDITIONUNKNOWN             *
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD ***
## LIGHTING_CONDITIONDAWN               *
## LIGHTING_CONDITIONDAYLIGHT
## LIGHTING_CONDITIONDUSK               **
## LIGHTING_CONDITIONUNKNOWN            ***
## ROADWAY_SURFACE_CONDICE
## ROADWAY_SURFACE_CONDOTHER
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT
## ROADWAY_SURFACE_CONDSNOW OR SLUSH    .
## ROADWAY_SURFACE_CONDUNKNOWN          **
## ROADWAY_SURFACE_CONDWET              **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 37662  on 50000  degrees of freedom
## Residual deviance: 37018  on 49980  degrees of freedom
## AIC: 37060
##
## Number of Fisher Scoring iterations: 10
```

Probit summary above

```
##
## Call:
## glm(formula = formula(best_model), family = binomial(link = "identity"),
##     data = df_sample, start = start_vals, control = glm.control(maxit = 100))
##
## Coefficients:
##                                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)                             0.0421271  0.0073778   5.710 1.13e-08
## POSTED_SPEED_LIMIT                      0.0021631  0.0001467  14.741  < 2e-16
## WEATHER_CONDITIONCLOUDY/OVERCAST        0.0285272  0.0092425   3.087  0.00203
## WEATHER_CONDITIONFOG/SMOKE/HAZE         0.0050905  0.0343300   0.148  0.88212
## WEATHER_CONDITIONOTHER                  0.0153338  0.0302207   0.507  0.61188
## WEATHER_CONDITIONRAIN                   0.0114861  0.0085738   1.340  0.18035
## WEATHER_CONDITIONSEVERE CROSS WIND GATE -0.0707915  0.0128066  -5.528 3.24e-08
## WEATHER_CONDITIONSLEET/HAIL             0.1372800  0.0488908   2.808  0.00499
## WEATHER_CONDITIONSNOW                   0.0025817  0.0105770   0.244  0.80716
## WEATHER_CONDITIONUNKNOWN               -0.0087136  0.0097079  -0.898  0.36941
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD 0.0444047  0.0069313   6.406 1.49e-10
## LIGHTING_CONDITIONDAWN                  0.0308999  0.0134236   2.302  0.02134
## LIGHTING_CONDITIONDAYLIGHT              0.0111545  0.0063292   1.762  0.07800
## LIGHTING_CONDITIONDUSK                  0.0289748  0.0106963   2.709  0.00675
## LIGHTING_CONDITIONUNKNOWN              -0.0172964  0.0097750  -1.769  0.07682
## ROADWAY_SURFACE_CONDICE                 0.0087843  0.0175287   0.501  0.61628
## ROADWAY_SURFACE_CONDOTHER               0.0031531  0.0283139   0.111  0.91133
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT     0.0342672  0.0727395   0.471  0.63757
## ROADWAY_SURFACE_CONDSNOW OR SLUSH      -0.0189307  0.0100794  -1.878  0.06036
## ROADWAY_SURFACE_CONDUNKNOWN            -0.0161170  0.0074947  -2.150  0.03152
## ROADWAY_SURFACE_CONDWET                 0.0222797  0.0071958   3.096  0.00196
##
## (Intercept)                             ***
## POSTED_SPEED_LIMIT                      ***
## WEATHER_CONDITIONCLOUDY/OVERCAST        **
## WEATHER_CONDITIONFOG/SMOKE/HAZE
## WEATHER_CONDITIONOTHER
## WEATHER_CONDITIONRAIN
## WEATHER_CONDITIONSEVERE CROSS WIND GATE  ***
## WEATHER_CONDITIONSLEET/HAIL             **
## WEATHER_CONDITIONSNOW
## WEATHER_CONDITIONUNKNOWN
## LIGHTING_CONDITIONDARKNESS, LIGHTED ROAD ***
## LIGHTING_CONDITIONDAWN                  *
## LIGHTING_CONDITIONDAYLIGHT              .
## LIGHTING_CONDITIONDUSK                  **
## LIGHTING_CONDITIONUNKNOWN               .
## ROADWAY_SURFACE_CONDICE
## ROADWAY_SURFACE_CONDOTHER
## ROADWAY_SURFACE_CONDSAND, MUD, DIRT
## ROADWAY_SURFACE_CONDSNOW OR SLUSH       .
## ROADWAY_SURFACE_CONDUNKNOWN             *
## ROADWAY_SURFACE_CONDWET                 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 37662  on 50000  degrees of freedom
## Residual deviance: 37101  on 49980  degrees of freedom
## AIC: 37143
##
## Number of Fisher Scoring iterations: 50
```

Identity model above.

Conclusion

In conclusion, this traffic collision study optimized a logistic regression model to isolate key predictors that drive vehicle crash severity in Chicago. Through evaluation, Logit and Probit link models were proven to be better compared to linear models, with the Logit model being the best fit due to the usefulness of Odds Ratios. Predictive threshold testing found a cut-off point of 13.82% that balances sensitivity and specificity. Moving forward, urban planners and emergency dispatchers can use models like these to deploy targeted public safety interventions and structural roadway modifications.